Applied Economics

040178-1 and 040178-2
Winter Term 2025/2026
Language: English
4 ECTS

University of Vienna

Instructor: Pello Aspuru
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Dates and Location

Course code 040178-1

- Wednesdays: 09:45–11:15, starting October 1, 2025.
- PC-Seminarraum 5, Oskar-Morgenstern-Platz 1, Floor: KG01, Room: U1.611

Course code 040178-2

- Thursdays: 16:45–18:15, starting October 2, 2025.
- PC-Seminarraum 5, Oskar-Morgenstern-Platz 1, Floor: KG01, Room: U1.611

Course Description

This is a course in which students will reinforce the tools learned in Introductory Econometrics (BA) by applying them to real-world data sets using econometric software. Prior knowledge at that level will be assumed throughout. The aim is for students to get hands-on experience analyzing observational data using the open-source programming language R and to acquire the tools to carry out empirical research projects independently.

Course Objectives

In this course you will learn to:

- Use econometric tools to produce empirical analyses.
- Interpret and critically evaluate existing empirical studies.
- Understand possibilities and limitations of empirical analyses.

Required Software

- Install R from `https://cran.r-project.org/`
- Install RStudio from `https://posit.co/download/rstudio-desktop/`

Course Schedule

This is a tentative and non-binding course schedule:

- Week 1 (Oct 14th / 15th): Introduction
- Week 2 (Oct 21st / 22nd): Data and Plotting

→ Deregistration period ends

- Week 3 (Oct 28th / 29th): Web Scraping Data
- Week 4 (Nov 4th / 5th): Multiple Linear Regression, Ordinary Least Squares (OLS) and Inference
- Week 5 (Nov 11th / 12th): Causality and Correlation
- Week 6 (Nov 18th / 19th): Panel Data

→ Week 7 (Nov 25th / 26th): Midterm exam

- Week 8 (Dec 2nd / 3rd): Instrumental Variables (IV)
- Week 9 (Dec 9th / 10th): Differences in Differences (DD)

→ Week 10 (Dec 16th / 17th): Project presentation I

Christmas break: no classes from Dec 23rd to Jan 7th

- Week 11 (Jan 13th / 14th): Regression Discontinuity Design (RDD)

→ Week 12 (Jan 20th / 21st): Project presentation II

→ Wednesday, Jan 27th: Final exam

Grading Policy

Item	**Tentative Date**	**Grade Weight**
Class Participation	Throughout the semester	10%
Midterm Exam	November 25th/26th 2026	25%
Project Presentation I	December 16th/17th 2026	10%
Project Presentation II	January 20th/21st 2027	15%
Final Exam	Wednesday, January 27th 2027	40%
Total		**100%**

Problem Set

- The problem set can be submitted in teams of up to three students.
- To register your problem set team:
  1. Enter this Google Sheet.
  2. Select your weekday tab.
  3. Add the team member names.

Academic Integrity

Students are expected to uphold the highest standards of academic honesty. Plagiarism or cheating will result in disciplinary action in accordance with university policy.

Accessibility

Students requiring accommodations should contact the instructor and the university's accessibility services.

Useful References

1. **Oswald** F, Viers V, Villedieu P, Kennedy G (2020). Introduction to Econometrics with R. SciencesPo Department of Economics, Paris, France.
   available online: https://scpoecon.github.io/ScPoEconometrics/.

2. **Heiss, F. (2020).** Using R for Introductory Econometrics (2nd edition)
   available online: http://urfie.net/readOnline.html

3. **Hanck**, C., Arnold, M., Gerber A., and Schmelzer M. (2023). Introduction to Econometrics with R
   available online: https://www.econometrics-with-r.org/index.html